

MODIFIED CAPSTONE PROJECT PROPOSAL

PRIIO: A Multi-Tenant QR-Based Digital Queue Management Platform

Consultation-ready concept paper aligned with the modified capstone proposal guidelines

Track Concentration: Web Applications Development

Primary Targeted SDG: SDG 9 - Industry, Innovation and Infrastructure

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Project Summary

Subject Area

Web application development for digital queueing, service operations, and customer experience management.

Aim

To build a tenant-aware queue platform that allows vendors to manage walk-in and remote customers through QR-based ticketing and live queue monitoring.

Core Argument

Many service providers still rely on manual or fragmented queueing methods that create crowding, uncertainty, and inconsistent queue visibility.

Expected Deliverable

A working web-based prototype with vendor dashboard, public queue board, QR join flow, and near-turn notification support.

Introduction

Service-oriented establishments such as clinics, school offices, retail counters, and small business service desks often struggle with long physical lines, manual ticket handling, and poor visibility into queue progress. Customers are forced to remain near the service area because they do not know when they will be called, while staff must manually manage the order of service and respond repeatedly to follow-up questions. This problem leads to crowded waiting areas, missed turns, inconsistent service flow, and unnecessary operational pressure on front-desk personnel.

Existing digital queueing solutions already attempt to improve this situation. Commercial systems such as Qminder, QLESS, and similar queue management platforms provide digital check-ins, service dashboards, and customer notifications. Some organizations also use appointment systems, messaging-based queue updates, or standalone queue display systems to reduce physical crowding. These solutions demonstrate that digital queueing can improve customer experience, especially when it gives users visibility into queue progress and reduces idle waiting time.

Despite these existing solutions, several gaps remain for the context targeted by this project. Many available platforms are subscription-based, optimized for larger organizations, or designed as closed systems that are difficult to adapt for smaller local service providers. Other solutions focus only on check-in or only on appointment booking, but do not combine QR-based entry, live queue monitoring, tenant-specific management, and configurable near-turn notifications in one lightweight web application. There is also a need for a system that allows multiple independent vendors to operate under one platform while still keeping their queue settings and ticket records separated.

The proposed solution is **Prio**, a multi-tenant QR-based digital queue management platform designed for small and medium-sized service environments. Prio gives each vendor its own workspace, public queue monitor, queue settings, and QR-based join link while allowing customers to join remotely or on site. The

system generates daily ticket numbers, provides live public queue updates, and supports email or SMS alerts when a customer is near the front or already called for service. By combining accessibility, tenant-aware design, and real-time queue visibility, the proposed platform directly addresses the identified gaps in existing queueing approaches.

Targeted SDG

Primary SDG: SDG 9 - Industry, Innovation and Infrastructure.

Prio supports SDG 9 by introducing a practical digital service infrastructure that helps organizations modernize how they manage queues and customer flow. The project applies web technologies to improve service efficiency, strengthen day-to-day operational processes, and provide an accessible innovation path for vendors that cannot easily invest in large enterprise systems.

Supporting alignment may also be argued for SDG 11 through reduced crowding in waiting areas and for SDG 8 through more efficient front-desk operations.

User Persona

Primary Persona: Vendor Queue Manager

Name: Maria Santos

Age: 38

Role: Receptionist / Small business front-desk staff

Needs: A faster way to issue queue numbers, monitor current and next tickets, and reduce repetitive customer inquiries about waiting time.

Pain Points: Manual ticket tracking, crowded waiting areas, missed turns, and difficulty notifying customers who step away from the service area.

Dashboard user

Needs live updates

Needs easy ticket controls

Secondary Persona: Queueing Customer

Name: Paolo Reyes

Age: 27

Role: Walk-in or remote customer seeking service

Needs: A simple way to join a queue, monitor progress remotely, and know when it is almost his turn.

Pain Points: Long idle waiting, uncertainty about queue position, and the risk of missing a turn when stepping away from the venue.

QR joiner

Needs queue visibility

Values notifications

Development Platform and Tools

Track / Platform	Web Applications Development
Frontend	React with Vite for a responsive browser-based interface
Backend	Node.js with Express for authentication, ticket management, and queue operations
Database	MongoDB with tenant-aware collections for users, vendors, tickets, and counters
Real-Time Updates	Server-Sent Events for live queue board and dashboard synchronization
Notifications	Email via SMTP and SMS via Twilio-compatible messaging workflow

Proposed Objectives

General Objective: To develop a multi-tenant digital queue management platform that improves service efficiency through QR-based joining, live queue monitoring, and near-turn customer notifications.

Specific Objectives:

- Enable vendor registration and tenant-specific queue workspace creation.
- Allow customers to join queues through QR codes or online join pages.
- Generate daily ticket numbers and lookup codes for queue tracking.
- Provide vendor controls for call next, serve current, skip, and walk-in issuance.
- Display a live public queue board accessible through a browser.
- Send email or SMS alerts when a customer is almost next or already called.

Proposed Features

Must-Have Features

- Vendor onboarding and tenant creation
- Customer account registration
- QR-based queue entry
- Online queue join page
- Daily ticket generation with queue prefixes
- Vendor dashboard for queue control
- Live public queue monitor
- Near-turn notifications through email or SMS

Nice-to-Have Features

- Multiple counters per tenant
- Business hours and queue availability controls
- Separate owner and staff roles
- Queue analytics and reporting
- Appointment windows or scheduled visits
- Rate limiting and CAPTCHA for public ticket creation

Feasibility for Proposal Consultation

The proposed system is feasible within the scope of a capstone project because its core workflow is clear and measurable: a customer joins a queue, receives a ticket, monitors queue movement, and is eventually called for service through a vendor-controlled dashboard. The selected technology stack is appropriate for a BSIT web development capstone because it requires applied work in interface design, backend API development, database modeling, real-time event handling, and external service integration.

The project also has a clear deliverable that can be demonstrated during consultation and defense: a working web prototype that shows vendor onboarding, QR queue entry, real-time monitoring, and queue state transitions. This makes the project suitable for evaluation in terms of technical implementation, usability, and problem-solution alignment.

Expected Consultation Talking Points

Problem Focus	Manual queueing causes crowding, uncertainty, and inefficient service handling.
Project Direction	Web-based queue management platform with tenant-aware service workspaces.
Clear Deliverable	Working prototype with vendor dashboard, public board, and QR join flow.

Innovation Angle	Combines multi-tenancy, QR self-service entry, live queue visibility, and alerting.
Evaluation Basis	Functionality, usability, queue accuracy, notification support, and feasibility.

End of modified capstone project proposal draft for individual consultation